

Detailed Forecast & Analysis Report

Predicted Load: 154.45936198811168 MWh

Uncertainty: ± 5.079434650833931 MWh

AI Executive Summary & Reasoning

The energy forecast for the specified period is projected to be **154.46 units**, with an associated uncertainty of **± 5.08 units**. This prediction is derived from an ensemble modeling approach, combining the outputs of three distinct machine learning algorithms: XGBoost, LightGBM, and CatBoost. CatBoost played the most significant role in the final forecast, contributing 65% of the weight, followed by LightGBM at 30%, and XGBoost at 5%. The individual model predictions ranged from 146.31 units (XGBoost) to 158.73 units (LightGBM), with the ensemble effectively balancing these diverse perspectives to arrive at the final robust estimate.

The primary driver behind this forecast is the **time of day**, specifically the 8 AM hour (on a Tuesday in January), which contributed a significant positive adjustment of +16.24 units to the baseline prediction. This suggests that energy demand is expected to be notably higher during this morning period, likely reflecting typical weekday operational ramp-up and activity. Historical consumption patterns also played a crucial role, with the energy consumption from 24 hours prior (lag_24h: 135 units) contributing +4 units, and consumption from 12 hours prior (lag_12h: 132.3 units) adding +2.19 units. These lagged features indicate a continuation of recent demand trends. Interestingly, **weather temperature had no discernible impact** on this particular forecast, suggesting other factors are dominant for this specific prediction window. The model effectively leverages these temporal and historical consumption insights to project the upcoming energy demand.

Key Factor Breakdown

- hour: 8
- day_of_week: 2
- month: 1
- is_weekend: 0
- lag_24h: 135
- lag_12h: 132.3
- roll_mean_24h: 137.7

- roll_std_24h: 5